

DUNCAN CAMERON-STEINKE

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Quantitative researcher and engineer completing an MSc in Mathematics and Statistics, researching uncertainty-aware deep learning and regime-switching volatility forecasting in equity markets. Three years in industry to senior data engineer, plus two years co-founding an analytics startup.

EXPERIENCE

SENIOR DATA ENGINEER

Aug 2022 – Apr 2024 1 yr 9 mos

ServiceTitan

- Embedded with executive, business development, and customer success teams to turn open business problems into shipped analytics, including a contract renewal dashboard that armed agents with product lift metrics ahead of renewal calls.
- Designed the data model and owned the pipelines behind a client-facing analytics product (Snowflake, SQL) serving 1,800+ customers.
- Designed the A/B testing methodology for an AI chat assistant automating call center support, using sample sizing and significance testing to measure impact on agent handling time.

ANALYTICS ENGINEER

Apr 2022 – Aug 2022 5 mos

Schedule Engine (acquired by ServiceTitan)

- Established the source of truth for reporting, defining business metrics where none existed and building the data analytics platform.

RESEARCH ENGINEER II, DATA SCIENCE

Jul 2021 – Feb 2022 8 mos

HomeX

- Deployed data ingestion microservices in Python and led client analytics projects.

CO-FOUNDER & CTO

Jun 2019 – Jul 2021 2 yrs 2 mos

Backr.ai (YouTube analytics startup)

- Co-founded a YouTube creator analytics platform, architecting a real-time streaming pipeline on GCP microservices in Python, with a React front end and a custom NLP model for comment classification and sentiment analysis.
- Won first prize (\$30K) at the DDQIC pitch competition and was selected for four accelerator programs including NextAI Montreal.
- Hired and led a team of three full-time developers and a data analyst, owning product roadmap and technical strategy from inception.
- Grew to 50 creator customers, including channels with 3M+ subscribers.

MACHINE LEARNING RESEARCH INTERN

Jan 2019 – Apr 2019 4 mos

Institute for Infocomm Research · Singapore

- Authored an NLP research paper (arXiv preprint) on unsupervised text expansion from pre-trained embeddings.
- Improved short-text classification accuracy under limited training data via pseudo-documents.

Research paper (arXiv) (<https://arxiv.org/abs/1909.07512>)

Singapore International Pre-Graduate Award

EDUCATION

THE UNIVERSITY OF BRITISH COLUMBIA

Sep 2024 – Aug 2026

Master of Science (MSc) · Kelowna, BC

Mathematics and Statistics

GPA: 4.0

- Coursework in non-parametric models, stochastic processes, differential equations, generalized linear models, and optimization.

QUEEN'S UNIVERSITY

Sep 2015 – May 2020

Bachelor of Applied Science (BASc) · Kingston, ON

Engineering Physics

GPA: 3.9

- Core studies in physics, mathematics, statistics, and computer programming.

Dean's Scholar · H. Jansen Memorial Scholarship (Awarded to the student with the highest standing in Physics I & II)

CERTIFICATIONS

MIT SCHWARZMAN COLLEGE OF COMPUTING

2022

Professional Certificate · Remote

Data Science and Machine Learning: Making Data-Driven Decisions

RESEARCH PROJECTS

BELIEF DISTRIBUTION LEARNING

Mar 2026 – Present

Research paper (in preparation)

- Created a novel deep learning algorithm grounded in belief theory by combining Subjective Logic with Evidential Deep Learning.
- Introduced per-label uncertainty modeling using Dirichlet distributions.
- Evaluated against published Label Distribution Learning algorithms, demonstrating improved accuracy and better-calibrated uncertainty estimates.

Python

TensorFlow

Keras

Machine Learning

Deep Learning

Uncertainty Quantification

VOLATILITY FORECASTING WITH UNCERTAINTY-AWARE COVARIATES

Jul 2025 – Apr 2026

Research project

- Added forecasting to the hidden Markov model package hmmTMB.
- Coupled an HMM with an uncertainty-aware deep learning model using a custom loss function.
- Demonstrated improved forecasting of volatility regimes in the TSX composite index compared to HMM or deep learning alone.

Python

R

Hidden Markov Models

Time Series Forecasting

Deep Learning

FORECASTING RADIO RECOMBINATION LINE SCIENCE

Sep 2025 – Jul 2026

MSc thesis, UBC Okanagan

- Scientific research project predicting the performance of a next-generation radio telescope.
- Extrapolated science results given the telescope design and existing surveys as priors.
- Optimized dish geometry to improve signal resolution.

Python

Math

Statistics

Astronomy

Research

SKILLS

A/B Testing · Analytics · Astronomy · AWS · Business Intelligence · Data Engineering · Data Pipelines · Deep Learning · Entrepreneurship · Full Stack · Hidden Markov Models · Keras · Leadership · Machine Learning · Math · NLP · Product Management · Production Software · Python · R · Research · Snowflake · SQL · Statistics · Tableau · Teamwork · TensorFlow · Time Series Forecasting · Uncertainty Quantification